FORMICA HYBRID ZONE · BAYPASS ASSOCIATION FOLLOW-UP

Ancestry-diagnostic enrichment in outlier regions, corrected for a cluster-size confound

A first pass suggested outlier regions from the PC1/PC2 BayPass scans are enriched for DIEM's ancestry-diagnostic markers. But the outlier definition mechanically favours large clusters, and cluster size alone predicts DiagnosticIndex — so we re-ran the comparison controlling for it. The corrected picture is more interesting than the first one.

Scope PC1 & PC2 · with/without Åland · with/without pre-estimated Ω **Prepared** 17 July 2026

Cluster size alone predicts DiagnosticIndex genome-wide (OR 1.14 per doubling, independent of any BayPass result), and outlier clusters are far larger than background by construction. Controlling for this, **PC1's enrichment survives in all 4 tested configurations** (adjusted OR 1.9–3.8×, still highly significant). **PC2's is fragile**: two configurations stay robustly enriched, but the other two *reverse sign* once size is accounted for — outlier clusters there have *fewer* diagnostic loci than equivalently large background regions, not more.

BACKGROUND

Why check this

DIEM assigns every marker a **DiagnosticIndex**: how strongly that locus distinguishes the two hybridizing lineages, independent of any covariate association test. A locus can only get into BayPass's outlier set by tracking PC1 or PC2 — DiagnosticIndex plays no part in that. So if outlier regions are enriched for high-DiagnosticIndex markers, that would be independent, orthogonal evidence that the regions BayPass is flagging are genuinely tied to hybrid ancestry, not statistical noise riding on population structure.

DiagnosticIndex > –25 is the same cutoff already used elsewhere in this pipeline to flag strongly diagnostic markers (`R/LD_decay_from_DIEM.R`). Genome-wide, only **4.5%** of markers clear it — most loci sit well below (median –49).

CONFOUND

Cluster size predicts DiagnosticIndex on its own

An **outlier cluster** is defined as an LD-based marker cluster (from the eMLG pipeline, `data/eMLG_5loci_0025.rds`) with ≥10 individually significant member loci (BF(dB) ≥ 20). That definition *requires* n_loci ≥ 10 — a cluster can't have 10 significant members without at least 10 members. So outlier clusters are mechanically biased toward the large end of the cluster-size distribution before BayPass results even enter the picture.

Separately, and independent of any BayPass run: larger LD clusters are themselves more likely to carry high-DiagnosticIndex loci (Spearman r=0.23 between cluster size and % high-DiagnosticIndex members across all 32,943 eMLG clusters; marker-level logistic regression OR=1.14 per doubling of cluster size, p≈0). That's biologically plausible on its own — larger non-recombining regions are exactly where multi-locus incompatibilities are expected to accumulate and persist — but it means part of any naive outlier-vs-background comparison could be picking up cluster size rather than anything about PC1/PC2 association specifically.

> A simple fix — restricting the background to clusters that are also ≥10 loci — turns out not to be enough. Checked directly on one configuration: outlier clusters there had a median of 147 loci (mean 743) against a median of just 15 (mean 28) for the size-floor-matched background. A shared floor does not mean a shared distribution.

METHOD

Naive test vs. size-adjusted test

- **Naive**: Fisher's exact test on the 2×2 table of (outlier vs. background) × (DiagnosticIndex > −25 vs. not) — the original comparison, with no size correction.
- **Size-adjusted**: a marker-level logistic regression, `DiagnosticIndex>−25 ~ is_outlier + log(cluster_size)`, fit separately per configuration. The coefficient on `is_outlier` is the enrichment that survives after cluster size is held constant continuously, rather than only truncated at a floor.

Both run separately for all 8 combinations of PC (1, 2) × population set (with Åland / Åland excluded) × Omega setting (pre-estimated from pruned data, or estimated internally).

RESULT

Naive vs. size-adjusted odds ratios

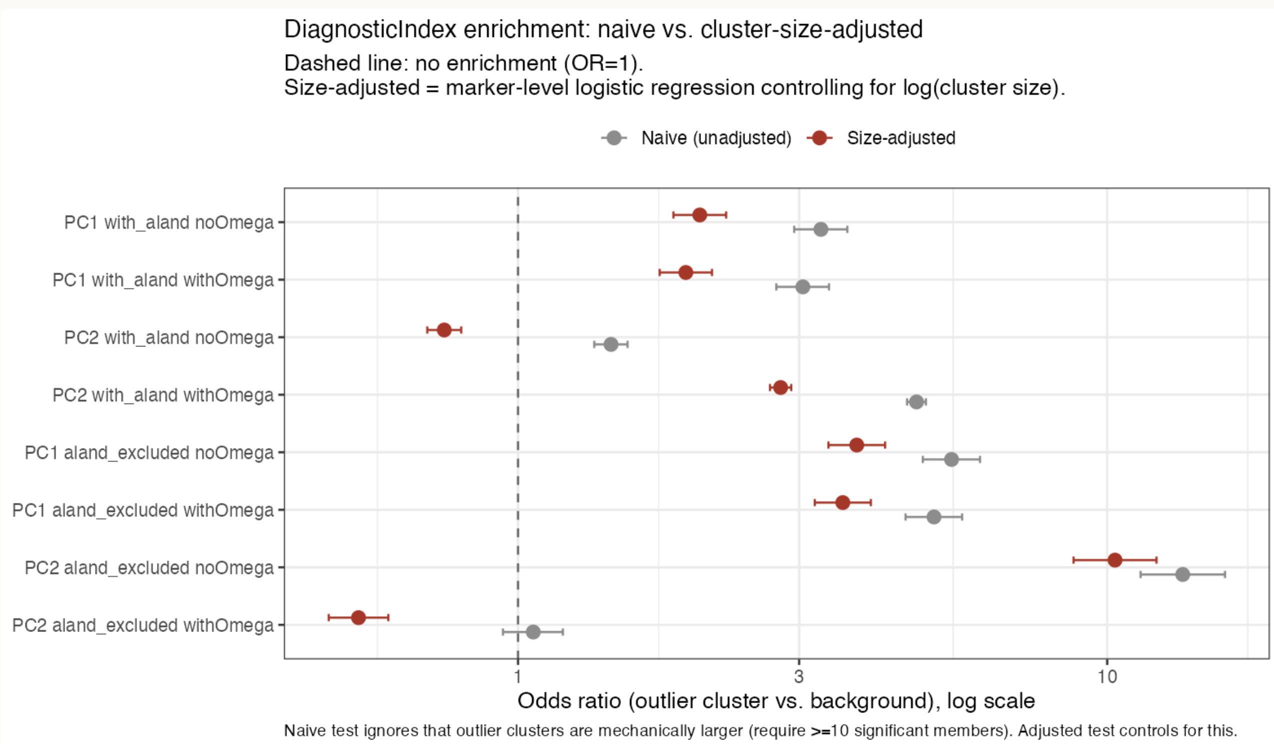

Figure 1. Odds ratio (outlier vs. background) per configuration, naive (grey) vs. size-adjusted (red), with 95% CIs. The gap between grey and red is exactly the part of the naive signal attributable to cluster size. Two PC2 configurations cross from >1 to <1 once adjusted.

Raw proportions (naive view)

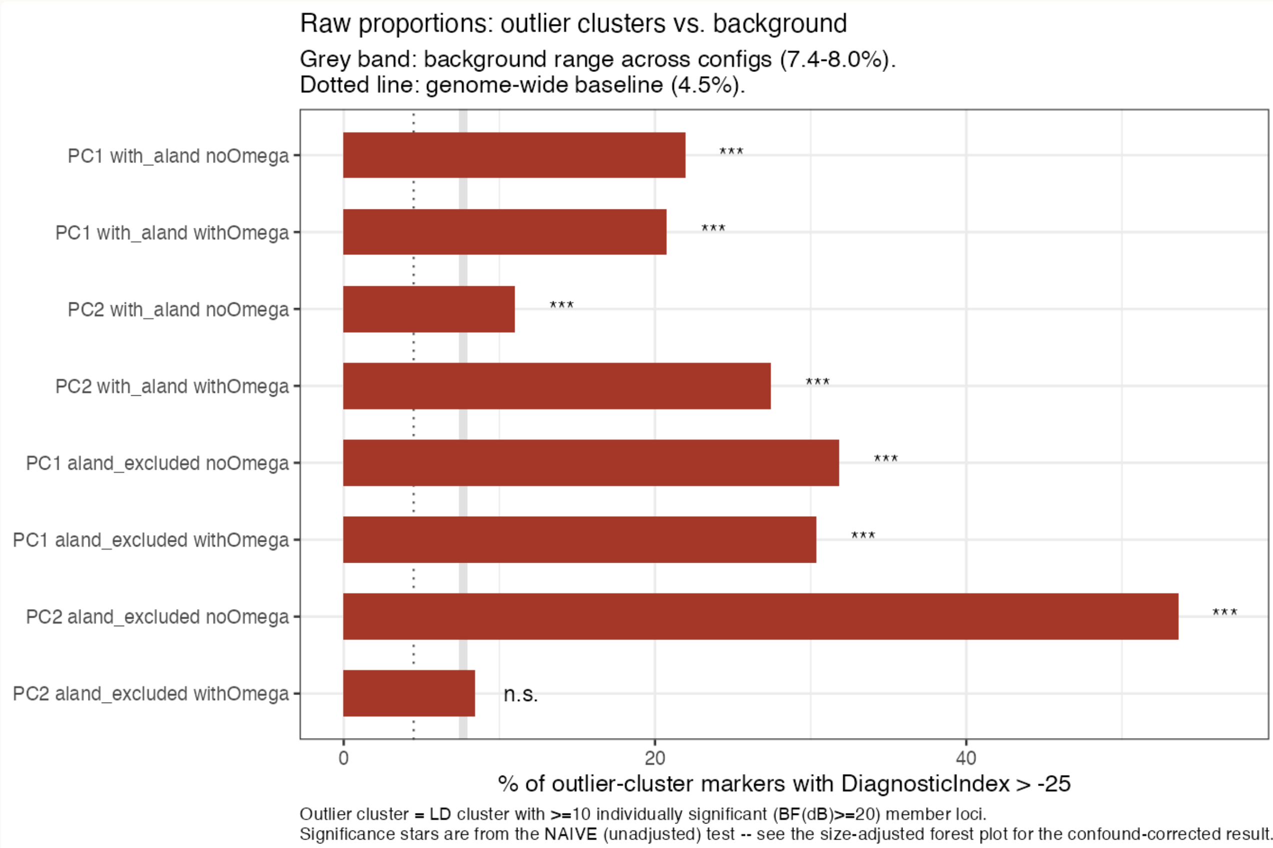

Figure 2. Percentage of outlier-cluster markers with DiagnosticIndex > −25, against the background range across all 8 configurations (grey band, 7.4–8.0%) and the genome-wide baseline (dotted line, 4.5%). Included for the plain-language view, but read alongside Figure 1 — these significance stars are from the naive test only.

Full results

PC	POPULATION SET	Ω SETTING	CLUSTERS	OUTLIER MARKERS	% OUTLIER	% BACKGROUND	NAI OR
PC1	with_aland	noOmega	3	2,161	21.9%	7.9%	3.27
PC1	with_aland	withOmega	4	2,284	20.8%	7.9%	3.04
PC2	with_aland	noOmega	13	9,664	11.0%	7.9%	1.44
PC2	with_aland	withOmega	20	15,787	27.5%	7.4%	4.75
PC1	aland_excluded	noOmega	5	1,469	31.9%	7.9%	5.44
PC1	aland_excluded	withOmega	6	1,540	30.4%	7.9%	5.08
PC2	aland_excluded	noOmega	3	593	53.6%	7.9%	13.4

PC	POPULATION SET	Ω SETTING	CLUSTERS	OUTLIER MARKERS	% OUTLIER	% BACKGROUND	NAI OR
PC2	aland_excluded	withOmega	4	3,776	8.4%	8.0%	1.06

INTERPRETATION

PC1 is consistent; PC2 splits in two

4 / 4
PC1 configurations still significant after adjustment, OR 1.9–3.8×

2 / 4
PC2 configurations still positively enriched, OR 2.8–10.3×

2 / 4
PC2 configurations reverse to significantly <1 after adjustment

PC1's enrichment survives correction in every setting, whether Åland is included or excluded and whether Omega is pre-estimated or left to BayPass. The adjusted odds ratios (1.9–3.8×) are smaller than the naive ones (3.0–5.4×) — part of the raw signal really was cluster size — but a large, consistent, highly significant effect remains in all four configurations. That stability across both nuisance variables is itself reassuring: it isn't an artifact of one particular modeling choice.

PC2 no longer reads as one consistent result. With Åland included and Omega pre-estimated (adjusted OR 2.79), and with Åland excluded and no pre-estimated Omega (adjusted OR 10.31, still the strongest result in the whole comparison), the enrichment holds up well after adjustment. But the other two PC2 settings — Åland included without pre-estimated Omega, and Åland excluded with it — **flip to significantly below 1** (adjusted OR 0.75 and 0.54, both p<1e-16). In those two configurations, once cluster size is accounted for, outlier clusters actually carry *fewer* ancestry-diagnostic loci than similarly large non-outlier regions — the opposite of the naive result, not just a weaker version of it.

> Practical takeaway: treat PC1 candidate regions as having independent support across the board. For PC2, the corrected result argues for checking *which specific configuration* a candidate region came from before treating it as validated — the Åland-included/no-pre-estimated-Omega and Åland-excluded/pre-

estimated-Omega settings no longer have this line of support, and the naive test alone would have missed that.

REPRODUCIBILITY

Code & data

Both figures and the results table are generated end-to-end by one script, reusing the same outlier-cluster definition as the Manhattan plots (`R/analyse_baypass.R`):

```
Rscript R/diagnostic_index_enrichment.R
```

This reads `data/hybrids_only_maf005.Rdata` (for DiagnosticIndex, one value per marker) and `data/eMLG_5loci_0025.rds` (LD clusters), plus the eight `*_summary_betai_reg.out` files under `with_aland/` and `aland_excluded/` , and writes:

- `data/diagnostic_index_enrichment.csv` — the full results table, naive and adjusted
- `Figures/diagnostic_index_enrichment_forest.png` — Figure 1
- `Figures/diagnostic_index_enrichment_proportions.png` — Figure 2

Size-adjusted test, per configuration

```
sig_ids <- dt[, .(n_sig = sum(BF >= 20, na.rm = TRUE)), by = group_id
             ][n_sig >= 10, group_id]
dt[, is_outlier := as.integer(group_id %in% sig_ids)]
dt[, hi := DiagnosticIndex > -25]

## naive: ignores cluster size entirely
fisher.test(table(dt$is_outlier, dt$hi))

## size-adjusted: the is_outlier coefficient is the enrichment that
## survives controlling for cluster size continuously
m <- glm(hi ~ is_outlier + log(n_loci), data = dt, family = binomial)
exp(coef(m)["is_outlier"])  # adjusted odds ratio
```